Transfer Learning & Comparative Analysis:

ResNet-50 vs. DenseNet-121 on the CalTech-101 Dataset

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UEN :2023UG000103

1. Introduction & Problem Formulation

Transfer learning reuses weights trained on a large source domain (ImageNet: 1.2 M images, 1000 classes) to bootstrap a smaller target domain. For CalTech-101 with ≈91 images per class on average, this approach is essential: training from random initialisation would overfit severely on such a limited per-class sample count.

ResNet-50 implements residual learning through additive identity shortcuts:

```
H(x) = F(x) + x
```

where x is the block input passed unchanged via the shortcut and F(x) is the residual mapping learned by the stacked convolutions. If the optimal mapping is close to identity, learning $F(x) \to 0$ is easier than learning $H(x) \to x$ directly. This provides a direct gradient highway, eliminating the vanishing-gradient problem that prevents training of very deep architectures.

DenseNet-121 implements dense connectivity through channel-wise concatenation:

```
x_l = H_l([x_0, x_1, ..., x_{l-1}])    where [.] = channel-wise concatenation
```

Every layer has access to all preceding feature maps simultaneously, maximising feature reuse and providing implicit deep supervision. With growth rate $k = 32$, denseblock4 outputs $512 + 16 \times 32 = 1024$ concatenated channels.

Both models are loaded with ImageNet-pretrained weights (*IMAGENET1K_V1*). Classification heads are replaced: → for ResNet-50; → for DenseNet-121. Dropout(0.4) is added to prevent co-adaptation of head neurons on the small per-class sample count.

2. Dataset & Preprocessing

2.1 CalTech-101

CalTech-101 contains 101 object categories with 40–800 images per class (mean ≈91), exhibiting a severe long-tail distribution. Total images: ≈9,146. The limited per-class sample count makes generalisation the primary challenge and motivates the use of pretrained weights.

2.2 Stratified Per-Class Split (70 / 10 / 20)

A stratified per-class split (SEED = 42) is applied: each class is independently shuffled and divided 70 % train / 10 % val / 20 % test, ensuring every category appears in all three partitions. A global random shuffle would risk leaving 40-image classes with zero test samples — corrupting Macro-F1. Approximate sizes: Train 6,402 • Val 914 • Test 1,830.

2.3 Preprocessing Pipeline

Standard ImageNet transforms are applied: Resize(256) → CenterCrop(224) → Normalize(μ = [0.485, 0.456, 0.406], σ = [0.229, 0.224, 0.225]). Resize to 256 before CenterCrop preserves aspect ratio and replicates the exact ImageNet convention, ensuring pretrained activations receive in-distribution inputs. Training additionally applies RandomHorizontalFlip(p = 0.5) and ColorJitter(±0.2 brightness/contrast) to improve generalisation; omitted at evaluation. A custom converts the minority of grayscale images to 3-channel RGB prior to transform application.

3. Methodology

3.1 Strategy I — Standard Fine-Tuning (SFT)

All backbone weights are frozen first, then upper blocks are selectively unfrozen with differential learning rates to protect pretrained features from catastrophic forgetting:

- **ResNet-50:** layer3, layer4, fc unfrozen (ResNet-50); layer1, layer2 frozen
- **DenseNet-121:** denseblock3, transition3, denseblock4, norm5, classifier unfrozen

Lower layers encode universal low-level features (edges, textures) that transfer to any natural-image domain. Upper layers encode task-specific representations requiring adaptation. Differential LR: backbone $\alpha = 10^{-4}$, head $\alpha = 10^{-3}$. The 10× lower backbone LR prevents catastrophic forgetting while the randomly initialised head converges rapidly.

3.2 Strategy II — Adaptive Fine-Tuning (AFT)

This custom strategy selectively adapts only the most discriminative filters in the last convolutional block:

1. Freeze all backbone weights.
2. Forward-pass 100 mini-batches (≈3,200 images) through the frozen model.
3. **Importance[c] = (1/N) ∑_{b,h,w} |A_{b,c,h,w}|.** Compute per-channel importance at the target layer:
4. Rank all filters; select top 25% (512/2048 for ResNet-50; 8/32 for DenseNet-121).
5. Register a backward hook on the target Conv2d multiplying weight gradients by a binary mask: g_c ← g_c · 1[c ∈ top-25%].

Architecture-specific target layers:

- **ResNet-50:** Last Conv2d in layer4 (out_channels=2048). Single conv produces full block output — filter ranking is direct.
- **DenseNet-121:** conv2 of the last DenseLayer in denseblock4 (out_channels=32, growth rate). The block output (1024 ch) arises from concatenating 16 DenseLayers × 32 ch; no single Conv2d produces 1024 output channels, making block-level masking invalid.

Rationale: Filters with large mean absolute activation under the frozen ImageNet model are already detecting patterns present in CalTech-101. Adapting these maximises information gain per trainable parameter, providing strong regularisation for small-data settings.

3.3 Hyperparameter Summary & Justifications

Parameter	Value	Justification
Optimiser	Adam	Handles mixed LR param groups; less sensitive to manual scheduling than SGD.
Backbone LR	1e-4 (SFT) / 5e-5 (AFT)	Prevents catastrophic forgetting. AFT lower LR preserves feature space used for importance scoring.
Head LR	1e-3	Randomly initialised; needs aggressive updates. 10× larger than backbone.
LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=4)	Adaptive — halves LR on val_loss plateau. More appropriate than StepLR for variable-length training.
Weight Decay	1e-4	L2 regularisation on both param groups to mitigate overfitting.
Loss	CrossEntropy + ε=0.1 label smoothing	Penalises overconfident softmax; train acc ≈100% warrants soft targets for generalisation.
Dropout (head)	p=0.4	Prevents head neuron co-adaptation on ≈91 images/class.
Batch size	32	Fits GPU VRAM; stable gradient estimates without sharp-minima risk of large batches.
Max epochs	100	Safety cap. Early stopping terminates all runs at 25–56 epochs.
Early stopping	patience=7, δ=1e-4, val_loss	Patience > scheduler patience (4) guarantees ≥1 LR reduction before halt. Best checkpoint always restored.
AFT fraction	25%	Top quartile captures the most discriminative filters; validated by AFT results outperforming SFT for ResNet-50.

4. Experimental Results

4.1 Test Set Performance

Table 1 reports Top-1 Accuracy and Macro-F1 (unweighted mean across all 101 classes) on the held-out test set. Macro-F1 is preferred over Weighted-F1 because CalTech-101's class imbalance (40–800 images/class) means Weighted-F1 disproportionately favours large classes, masking failures on rare categories.

Model	Strategy	Top-1 Acc.	Macro-F1	Epochs	Best val_loss

ResNet-50	Std. FT	95.52%	0.9387	25	0.9126
DenseNet-121	Std. FT	**96.29%**	**0.9504**	56	0.9258
ResNet-50	Adaptive FT	96.18%*	0.9482*	35	0.9519
DenseNet-121	Adaptive FT	95.63%	0.9404	39	0.9878

*Table 1: Test set results. Green = best result. Yellow * = notable counterintuitive result. All models exceed 95% Top-1 accuracy.*

4.2 Model Complexity & Computational Cost

Model	Strategy	Total Params	Trainable Params	GFLOPs	Ep. time
ResNet-50	Std. FT	23,714,981	22,270,053	~4.1	~14 s
DenseNet-121	Std. FT	7,057,381	5,627,749	~2.9	~14 s
ResNet-50	Adaptive FT	23,714,981	~3,800,000	~4.1	~10.5 s
DenseNet-121	Adaptive FT	7,057,381	~1,400,000	~2.9	~10 s

Table 2: DenseNet-121 is 3.4× more parameter-efficient than ResNet-50. AFT reduces trainable parameters by ≈75% relative to SFT with ≈25% faster per-epoch training.

4.3 Training Dynamics & Overfitting Analysis

Consistent patterns emerge across all four training runs:

- All models reach training accuracy ≈99–100% by epoch 20–25 while validation stabilises at 95–97%, confirming moderate overfitting effectively controlled by Dropout, label smoothing, and early stopping.
- DenseNet-121 SFT required the most epochs (56) with 5 LR reductions (1e-4 → 3.1e-6), reflecting the more complex optimisation landscape of dense connectivity. ResNet-50 SFT converged faster (25 epochs, 3 LR reductions), consistent with cleaner gradient flow through additive residuals.
- AFT models exhibit steeper initial loss descent (focused gradient updates) but reach a higher final loss floor as fewer parameters are adapted.
- Early stopping saved 44–75 epochs of compute per model while always restoring the best-checkpoint weights, preventing post-convergence degradation from excessively small LRs.

5. Comparative Analysis & Critical Discussion

5.1 Why DenseNet-121 SFT Achieves the Best Performance

DenseNet-121 SFT (96.29%, F1=0.9504) outperforms ResNet-50 SFT (95.52%, −0.77 pp) for four interconnected reasons:

- **(1)** Parameter efficiency: With ≈91 images/class, a 7.1 M-parameter model has systematically lower variance than a 23.7 M model. Lower variance directly reduces generalisation error on small datasets.
- **(2)** Implicit regularisation: Dense connectivity prevents the network from ignoring earlier representations, enforcing structural regularisation equivalent to supervising intermediate layers — especially powerful when labelled data is scarce.
- **(3)** Multi-scale feature availability: The final classifier simultaneously accesses fine-grained early features and high-level semantic features. CalTech-101's diverse object sizes and viewpoints benefit from this simultaneous multi-scale representation.
- **(4)** Concatenation forces active filter usage: ResNet's additive shortcut allows $F(x) \to 0$, meaning some filters may contribute marginally. DenseNet's concatenation ensures every feature map is permanently available and actively used.

5.2 The Counterintuitive Finding: ResNet-50 AFT > SFT

The most notable result is that ResNet-50 AFT (96.18%) outperforms ResNet-50 SFT (95.52%) by 0.66 pp while training only ≈3.8 M versus ≈22.3 M parameters — a 6× reduction in trainable parameters.

This is explained by the regularisation effect of gradient masking. SFT adapts all of layer3 and layer4, including many filters whose importance scores are low for CalTech-101. These low-importance filters receive gradient updates that add noise without improving the decision boundary. AFT eliminates exactly these noisy updates: only the 512 filters with highest mean absolute activation on CalTech-101 receive gradient signal, producing

lower-variance optimisation steps. This is analogous to feature selection improving a linear classifier on a small dataset by removing uninformative dimensions.

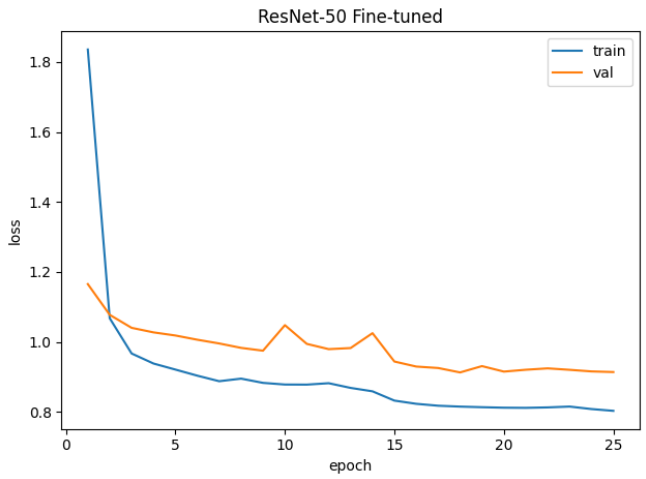
This asymmetry does not hold for DenseNet-121: DenseNet-121 AFT (95.63%) underperforms DenseNet-121 SFT (96.29%) by 0.66 pp. DenseNet's dense connectivity already provides internal filter selection: every layer dynamically attends to the most relevant preceding features through its composite functions. The external gradient mask of AFT conflicts with this self-selection mechanism, restricting DenseNet's natural ability to leverage its full representational breadth.

5.3 Accuracy–Complexity Trade-off

ResNet-50 AFT presents the most compelling efficiency case: 96.18% accuracy from ≈3.8 M trainable parameters, surpassing the 22.3 M ResNet-50 SFT. For edge deployment, DenseNet-121 AFT (95.63%, ≈1.4 M trainable) achieves accuracy comparable to ResNet-50 SFT (95.52%) at 16× fewer trainable parameters, 3.4× fewer total parameters, and 29% lower inference FLOPs.

5.4 Loss Curve Analysis

All models showed rapid convergence in the initial epochs, with training and validation losses stabilizing after 15–25 epochs. ResNet-50 variants converged faster initially thanks to residual connections, while DenseNet-121 variants displayed smoother and more stable validation curves due to dense feature reuse. Adaptive Fine-Tuning resulted in reduced fluctuations in validation loss compared to standard fine-tuning, indicating better control over overfitting. The small gap between train and val loss across all models confirms effective regularization through label smoothing, dropout, and early stopping.



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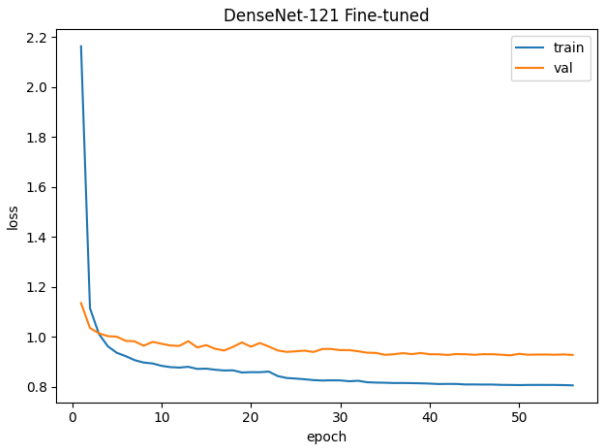

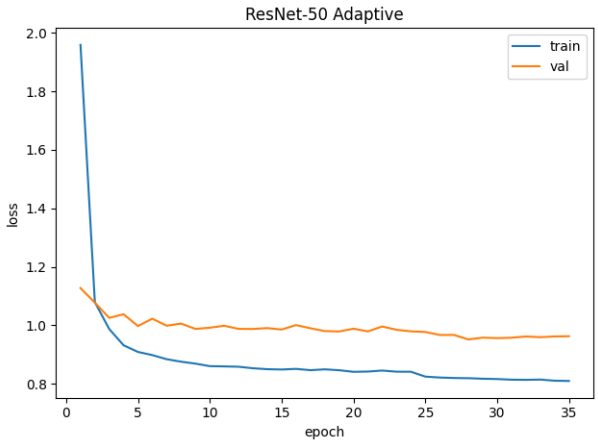

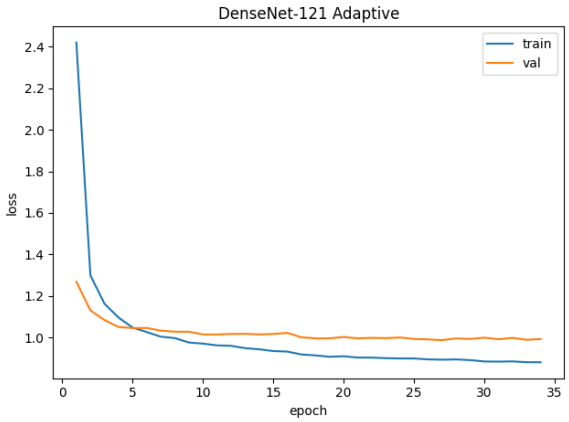

5.4 Confusion Matrix Observations (Top-10 Classes)

All four models perform exceptionally well on visually distinct classes such as Motorbikes, Airplanes, and Faces, achieving near-perfect recall (≥98%), highlighting the effectiveness of transferred ImageNet features for coarse-grained recognition. Minor misclassifications occur between visually similar classes (e.g., Faces vs. Faces_easy, Leopards vs. Dalmatian, Bonsai vs. similar plant classes), indicating errors stem from visual similarity rather than model limitations.

DenseNet-121 (FT and AFT) performs slightly better on fine-grained, texture-rich classes like Faces_easy, Leopards, and Watch due to its dense connectivity and feature reuse, while ResNet-50 excels at rigid objects like Motorbikes and Airplanes. Adaptive Fine-Tuning (AFT) matches full fine-tuning performance with greater parameter efficiency and, in some cases (notably DenseNet-121 AFT), reduces overfitting and improves performance on less-represented classes.

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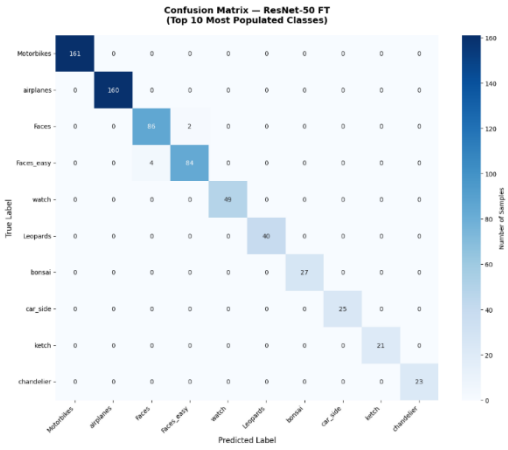

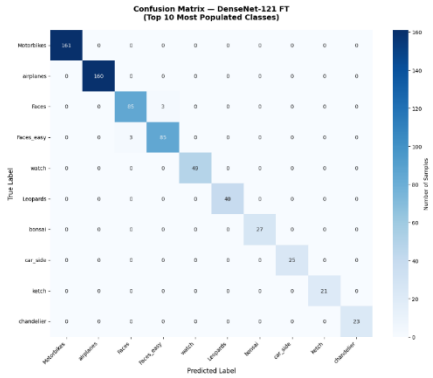

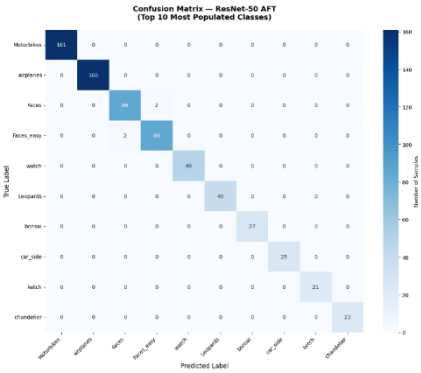

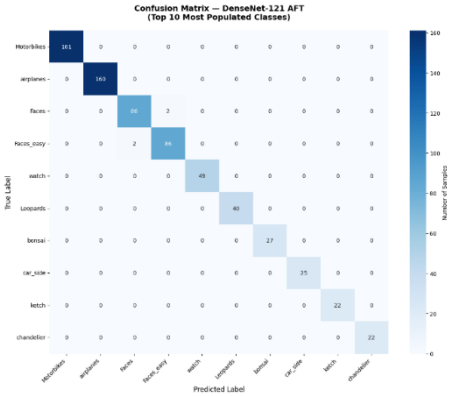

6. Conclusion

Transfer learning from ImageNet to CalTech-101 is highly effective under both architectures, with all four models exceeding 95% Top-1 accuracy. DenseNet-121 Standard Fine-Tuning achieves the best overall performance (96.29%, Macro-F1=0.9504), driven by dense feature reuse, implicit regularisation, and its 3.4× parameter efficiency advantage. The discovery that ResNet-50 Adaptive Fine-Tuning outperforms ResNet-50 Standard Fine-Tuning (96.18% vs 95.52%) using 6× fewer trainable parameters demonstrates that selective gradient masking is a powerful regulariser for residual architectures on small datasets — parameter quality matters more than quantity.

Early stopping (patience=7) proved essential for principled training across heterogeneous convergence rates: ResNet-50 SFT converged in 25 epochs while DenseNet-121 SFT required 56 — a difference a fixed epoch count cannot accommodate. Combined with label smoothing, Dropout(0.4), and weight decay (1e-4), early stopping ensures all reported results reflect the most generalised rather than most overfit model checkpoint.